

Question	Marking Guidance	Mark	Comments
05.1	Accept suitable null hypothesis that includes type of light and behaviour, eg The type of light has no effect on the behaviour/movement of COTS OR There is no difference in behaviour/movement with constant/flashing light;	1	Ignore general null hypotheses, or example 'there is no difference between observed and expected'
05.2	Accept any two factors for one mark from the list below; Salinity / salt concentration of the water Temperature (of the water) Amount / distribution of food pH (of the water) Oxygen/carbon dioxide concentration Intensity/wavelength of (constant and flashing) light	1 max	List rule applies Ignore humidity Ignore type of coral Ignore depth of water
05.3	Yes (no mark) 1. Movement is away from either type/both types of light OR Negative (photo) taxis to both types/either types of light; 2. Significant movement <u>away from</u> constant light as $p=0.02/<0.05/=2\%/<5\%$ OR Movement <u>away from</u> constant light is not due to chance as $p=0.02/<0.05/=2\%/<5\%$; No (no mark) 3. Movement away from flashing light is not significant as $p=0.69/>0.05/=69\%/>5\%$ OR Movement away from flashing light is due to chance as $p=0.69/>0.05/=69\%/>5\%$;	3	2. and 3. Ignore ' results ' in the context of significance or chance

05.4	Correct answer of 3 hours = 2 marks;; Allow 1 mark for distance of 48 000 mm in working 1 max for answer of 185 minutes/3 hours and 5 minutes/3.09 hours 1 max for answer of 1 hour (ie answers that use 564 in their calculation);	2 max	
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